🗄️ Supabase Setup Guide

Complete step-by-step guide to set up Supabase database for your AI Feedback System.

🎯 Why Supabase?

✅ **Persistent Storage** - Data survives app restarts
✅ **Real-time Updates** - Perfect for Admin Dashboard
✅ **SQL Queries** - Powerful analytics and filtering
✅ **Free Tier** - 500 MB database, 2 GB bandwidth/month
✅ **PostgreSQL** - Industry-standard, production-ready
✅ **Easy Integration** - Simple Python client

📋 Step-by-Step Setup

Step 1: Create Supabase Account

1. **Go to Supabase**
   - Visit: https://supabase.com
   - Click "Start your project"
2. **Sign Up**
   - Use GitHub, Google, or email
   - Verify your email
3. **Create Organization**
   - Name: Your name or project name
   - Click "Create organization"

Step 2: Create New Project

1. **New Project**
   - Click "New project"
   - Choose your organization

2. **Project Settings**

```
Name: ai-feedback-system (or any name)
Database Password: [Create a strong password]
Region: Choose closest to you (e.g., US East, Mumbai)
Pricing Plan: Free
```

3. **Create Project**
    - Click "Create new project"
    - Wait 2-3 minutes for setup

Step 3: Create Database Table

1. **Open SQL Editor**
    - In your project dashboard
    - Left sidebar → "SQL Editor"
    - Click "New query"
2. **Run This SQL**

```sql
```

```sql
-- Create feedback table
CREATE TABLE feedback (
  id BIGSERIAL PRIMARY KEY,
  timestamp TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  rating INTEGER NOT NULL CHECK (rating >= 1 AND rating <= 5),
  review TEXT NOT NULL,
  ai_response TEXT NOT NULL,
  summary TEXT NOT NULL,
  recommended_actions TEXT[] NOT NULL,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- Create index for faster queries
CREATE INDEX idx_feedback_rating ON feedback(rating);
CREATE INDEX idx_feedback_timestamp ON feedback(timestamp DESC);

-- Enable Row Level Security (RLS)
ALTER TABLE feedback ENABLE ROW LEVEL SECURITY;

-- Create policy to allow all operations (for now)
CREATE POLICY "Allow all operations" ON feedback
  FOR ALL
  USING (true)
  WITH CHECK (true);
```

3. **Execute**
   - Click "Run" (or Ctrl/Cmd + Enter)
   - You should see: "Success. No rows returned"
4. **Verify Table**
   - Left sidebar → "Table Editor"
   - You should see "feedback" table
   - Columns: id, timestamp, rating, review, ai_response, summary, recommended_actions, created_at

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Step 4: Get API Credentials

1. **Open Settings**
   - Left sidebar → "Project Settings"

- Click "API"

2. **Copy These Values Project URL:**

```
https://abcdefghijk.supabase.co
```

⚠️ Copy your actual URL!

Anon/Public Key:

```
eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

⚠️ Copy the FULL key (it's very long!)

3. **Keep These Safe**
   - Don't share publicly
   - We'll add them to `.env` file

Step 5: Add to Your Project

1. **Update `.env` File**

```bash
# Open your .env file
nano .env

# Add these lines (replace with your actual values):
GEMINI_API_KEY=your_gemini_key_here
SUPABASE_URL=https://your-project-id.supabase.co
SUPABASE_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...

# Save and close (Ctrl+X, Y, Enter)
```

2. **Test Connection**

```bash
python supabase_client.py
```

Expected output:

```
✅ Supabase connection initialized!
✅ Supabase table access successful!
✅ Save successful!
```

🧪 Testing Your Setup

Test 1: Connection Test

```python
from supabase_client import test_supabase_connection

if test_supabase_connection():
    print("✅ Everything works!")
else:
    print("❌ Check your credentials")
```

Test 2: Save Test Data

```python
from supabase_client import save_feedback_to_db
from datetime import datetime

success = save_feedback_to_db(
    timestamp=datetime.now().isoformat(),
    rating=5,
    review="Test review",
    ai_response="Test response",
    summary="Test summary",
    actions=["Action 1", "Action 2"]
)

print(f"Save successful: {success}")
```

Test 3: Load Data

```python
```

```
from supabase_client import load_recent_feedback_from_db

recent = load_recent_feedback_from_db(5)
print(f"Loaded {len(recent)} entries")
for entry in recent:
    print(f"- {entry['rating']} ⭐: {entry['review'][:50]}...")
```

🔐 Security Best Practices

✅ DO:

- Use environment variables for credentials
- Keep `.env` in `.gitignore`
- Use RLS (Row Level Security) policies
- Use `anon` key for public access (it's safe!)

❌ DON'T:

- Hardcode credentials in code
- Commit `.env` to GitHub
- Share your service_role key publicly
- Disable RLS without policies

📊 Database Schema Explained

Table: `feedback`

Column	Type	Description
`id`	BIGSERIAL	Auto-incrementing primary key
`timestamp`	TIMESTAMPTZ	When feedback was submitted
`rating`	INTEGER	1-5 star rating (validated)
`review`	TEXT	User's review text

Column	Type	Description
`ai_response`	TEXT	AI-generated user response
`summary`	TEXT	AI-generated summary
`recommended_actions`	TEXT[]	Array of action items
`created_at`	TIMESTAMPTZ	Row creation time

Indexes

- `idx_feedback_rating` - Fast filtering by rating
- `idx_feedback_timestamp` - Fast sorting by date

Constraints

- `rating` must be between 1-5
- All text fields are NOT NULL

🚀 Using in Your App

Automatic Integration

Your app now automatically:

1. ✅ Tries Supabase first
2. ✅ Falls back to CSV if Supabase unavailable
3. ✅ Saves to both (dual storage)

Check Storage Mode

```python
```

```
from utils import USE_SUPABASE

if USE_SUPABASE:
    print("Using Supabase")
else:
    print("Using CSV fallback")
```

Manual Operations

```python
# Save feedback
from supabase_client import save_feedback_to_db
save_feedback_to_db(...)

# Load recent
from supabase_client import load_recent_feedback_from_db
data = load_recent_feedback_from_db(10)

# Get statistics
from supabase_client import get_feedback_stats_from_db
stats = get_feedback_stats_from_db()

# Search by keyword
from supabase_client import search_feedback
results = search_feedback("food")

# Filter by rating
from supabase_client import get_feedback_by_rating
five_stars = get_feedback_by_rating(5)
```

🔧 Troubleshooting

Issue 1: "Connection failed"

Cause: Wrong credentials or network issue

Solution:

```bash
```

```
# Check credentials
python -c "import os; from dotenv import load_dotenv; load_dotenv(); print(os.getenv('SUPABASE_URL'))"

# Should print your Supabase URL
# If it prints None, your .env is not loaded correctly
```

Issue 2: "Table 'feedback' does not exist"

Cause: Table not created yet

Solution:

1. Go to Supabase Dashboard → SQL Editor
2. Run the CREATE TABLE SQL from Step 3 above
3. Verify in Table Editor

Issue 3: "Permission denied"

Cause: RLS policy not configured

Solution:

```sql
-- In SQL Editor, run:
CREATE POLICY "Allow all operations" ON feedback
   FOR ALL
   USING (true)
   WITH CHECK (true);
```

Issue 4: "Array type not supported"

Cause: Trying to use lists directly

Solution: Supabase handles arrays natively - just pass Python lists!

```python
actions = ["Action 1", "Action 2"]  # ✅ This works!
```

📈 Monitoring Your Database

View Data in Supabase

1. **Table Editor**
    - See all your feedback entries
    - Filter, sort, search
    - Edit or delete rows
2. **Database Stats**
    - Dashboard → "Database"
    - See storage used
    - Monitor queries
3. **Logs**
    - Dashboard → "Logs"
    - See all API calls
    - Debug issues

💰 Free Tier Limits

What you get FREE forever:

- ✅ 500 MB database storage
- ✅ 2 GB bandwidth per month
- ✅ 50,000 monthly active users
- ✅ 500 MB file storage
- ✅ 2 GB egress

For this assignment: More than enough!

Upgrade only if: You get 1000+ users/month

🎯 For Streamlit Cloud Deployment

Add secrets in Streamlit Cloud:

```toml
# In Streamlit Dashboard → Settings → Secrets

GEMINI_API_KEY = "your_gemini_key"
SUPABASE_URL = "https://your-project.supabase.co"
SUPABASE_KEY = "your_anon_key"
```

📚 Additional Resources

- **Supabase Docs**: https://supabase.com/docs
- **Python Client**: https://supabase.com/docs/reference/python
- **SQL Tutorial**: https://supabase.com/docs/guides/database

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✅ Verification Checklist

Before proceeding, ensure:

- [ ] Supabase account created
- [ ] Project created
- [ ] `feedback` table created
- [ ] API credentials copied
- [ ] Credentials added to `.env`
- [ ] `python supabase_client.py` runs successfully
- [ ] Test data saved successfully
- [ ] Can view data in Supabase Table Editor

🎉 You're Done!

Your database is now:

- ✅ **Live** and accessible

- ✅ **Integrated** with your app
- ✅ **Production-ready**
- ✅ **Free forever** (within limits)

Run your app and start collecting feedback! 🚀

```bash
streamlit run app.py
```

Every submission now saves to Supabase! 🎊